Appendix A – Measured raw data

The general geophysical data of all samples that have been tested. Data are displayed by three lithofacies. For each sample, the routine helium porosity and air permeability with Klinkenberg-correction (K_{air}) from special core analysis, NMR estimated porosity, characteristic T_2 value T_{2ML}, signal to noise ratio (SNR) of NMR test, real (σ') and imaginary (σ'') part of conductivity at 1Hz, recorded frequency range of SIP test are shown.

Lithofacies	Sample I.D.	Helium Porosity	NMR Porosity	Air Permeability	T_{2ml}	NMR SNR	Real conductivity (f=1Hz)	Imaginary conductivity (f=1Hz)	Frequency range
		(%)	(%)	(mD)	(ms)		(S/m)	(S/m)	(Hz)
Grainstones	3	22.77	20.22	3.822	270.291	154	2.12E-03	1.13E-05	0.1-10k
	9	18.10	15.26	0.718	174.473	134.3	2.47E-03	2.26E-06	0.01-10k
	11	20.31	19.33	14.910	278.690	161.1	2.69E-03	7.10E-06	0.01-10k
	16	26.40	25.30	1.137	660.808	183	2.57E-03	3.25E-06	0.1-10k
	27	10.76	6.91	0.020	198.001	53.5	1.22E-03	1.55E-06	0.01-10k
	37	19.99	20.97	0.338	520.498	210.7	2.11E-03	2.80E-06	0.01-10k
	38	24.55	16.60	0.128	196.411	193.2	2.57E-03	1.82E-06	0.01-10k
	39	23.87	25.30	1134.267	248.913	98	9.21E-03	1.20E-05	0.01-10k
	44	21.52	20.00	15.879	459.828	76.5	5.80E-03	7.28E-06	0.01-10k
	51	7.53	9.21	2.806	103.599	109.4	1.51E-03	4.26E-06	0.01-10k
	51b	5.98	6.40	0.012	68.540	76.8	4.87E-04	2.16E-06	0.01-10k
	53	14.00	16.39	19.255	218.479	183.3	2.42E-03	1.79E-05	0.01-10k
	57	20.11	21.28	3.447	279.984	186	3.31E-03	3.81E-06	0.01-10k
	59	10.02	6.40	0.094	33.270	57.3	2.61E-03	3.08E-06	0.01-10k
	64	26.68	26.99	15.891	467.006	228.2	4.27E-03	4.14E-06	0.1-10k
	65	20.01	19.75	4.040	245.942	171.5	3.45E-03	6.42E-06	0.1-10k
	66	21.02	19.92	3.906	236.212	178.6	3.04E-03	2.89E-06	0.01-10k
	67	19.82	17.72	6.188	202.791	157.6	3.35E-03	5.49E-06	0.01-10k
	69	2.46	2.81	0.006	14.004	14.9	8.30E-04	3.24E-06	0.01-10k
	70	8.68	9.20	2.112	336.761	95.6	1.32E-03	7.83E-06	0.1-10k
	71	4.15	4.17	0.002	33.256	27.3	1.04E-03	1.49E-06	0.01-10k
	72	7.08	7.40	0.230	39.572	86.4	1.28E-03	8.75E-06	0.1-10k
	81	25.58	26.26	50.360	331.364	227.9	6.43E-03	3.35E-06	0.1-10k
	82	18.41	16.87	10.730	218.106	199.6	3.54E-03	2.20E-06	0.01-10k

	85	24.08	24.35	22.369	350.604	283.2	4.46E-03	2.83E-06	0.01-10k
	86	23.18	23.39	15.090	340.390	244.8	5.86E-03	2.71E-06	0.1-10k
	89	5.73	6.70	0.010	10.942	66.3	6.95E-04	9.21E-06	0.1-10k
	92	32.17	33.50	11.856	672.518	203.8	2.74E-03	4.18E-06	0.01-10k
	101	25.32	24.78	5.076	287.135	253.9	3.71E-03	9.31E-06	0.01-10k
	109b	22.46	22.50	22.538	440.224	226.9	3.29E-03	3.66E-06	0.01-10k
	115	32.60	32.10	16.121	844.290	324.5	3.62E-03	5.65E-06	0.01-10k
	123	20.36	20.80	0.649	288.055	149.8	2.36E-03	4.70E-06	0.01-10k
	130	19.71	12.42	0.155	189.025	90.1	3.06E-03	2.16E-06	0.01-10k
	135	8.28	6.96	0.031	48.486	60.6	1.11E-03	1.28E-06	0.01-10k
	145	5.28	5.84	0.036	17.405	40.9	1.36E-03	7.49E-06	0.1-10k
	151	20.46	20.97	3.462	326.521	152	2.41E-03	4.08E-06	0.01-10k
	154	15.05	14.90	0.169	363.482	118.6	1.48E-03	3.87E-06	0.1-10k
	156	3.09	3.99	0.021	51.515	30.1	1.06E-03	2.37E-06	0.01-10k
	157	5.11	7.03	0.073	55.416	64.2	3.77E-03	5.68E-05	0.01-10k
	164	24.53	25.94	0.309	418.273	267	2.71E-03	1.82E-06	0.01-10k
	165	20.27	22.08	0.971	373.761	244.2	3.19E-03	7.12E-06	0.1-10k
Packstones	13-18	7.70	6.10	1.970	109.049	52.6	9.58E-03	3.19E-05	0.01-10k
	13-46	2.60	3.90	0.001	171.963	106.8	2.59E-03	1.68E-06	0.01-10k
	14-9	12.70	8.30	239.130	124.941	114.1	3.72E-03	1.64E-05	0.01-10k
	18-30	4.70	4.30	293.490	99.231	61.1	7.50E-03	1.45E-05	0.01-10k
	18-33	5.30	5.00	0.010	101.810	69.5	5.72E-03	1.36E-05	0.01-10k
Mudstones	16-15	3.20	5.70	0.001	37.740	95.9	7.32E-03	4.24E-05	0.01-10k
	20-37	1.70	1.80	297.160	35.793	18.7	2.96E-03	6.87E-06	0.01-10k
	30-4	3.40	3.70	0.001	29.825	47.7	2.36E-03	1.55E-05	0.01-10k
	30-24	2.80	2.70	0.001	24.597	65.6	2.01E-03	1.25E-05	0.01-10k
	33-25	11.80	11.60	0.100	225.979	108.5	1.58E-02	3.80E-05	0.01-10k

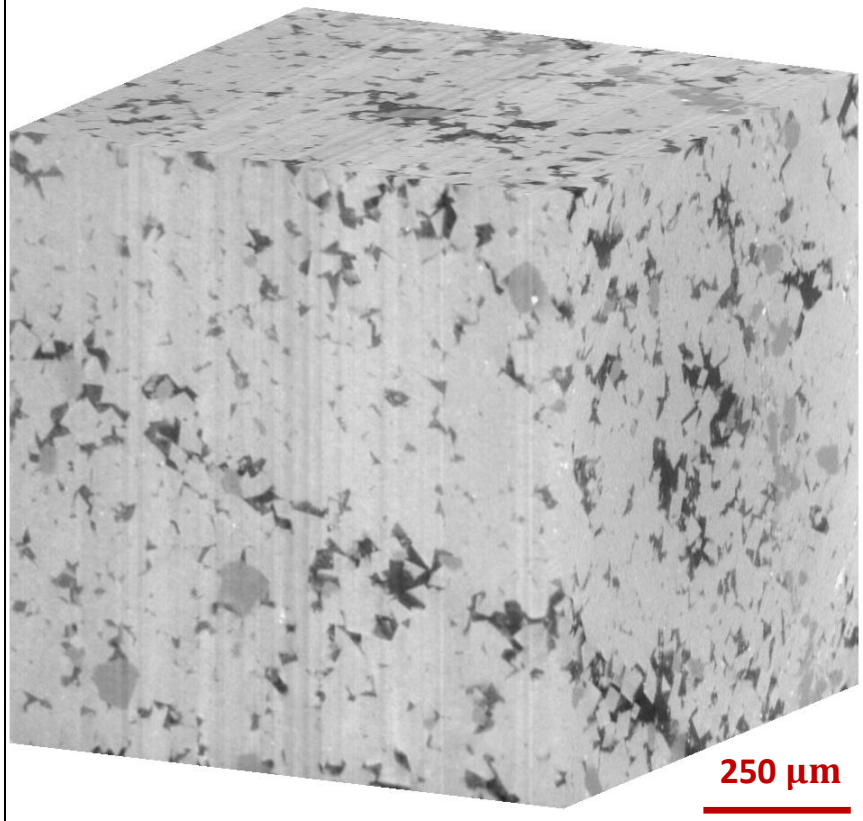
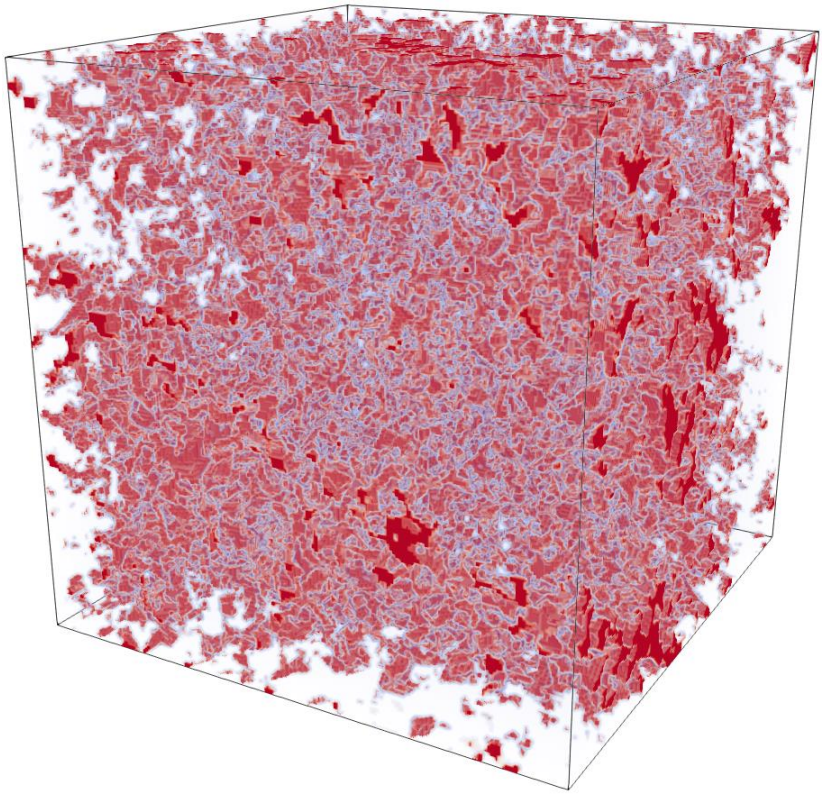
Appendix B – Calculated petrophysical properties

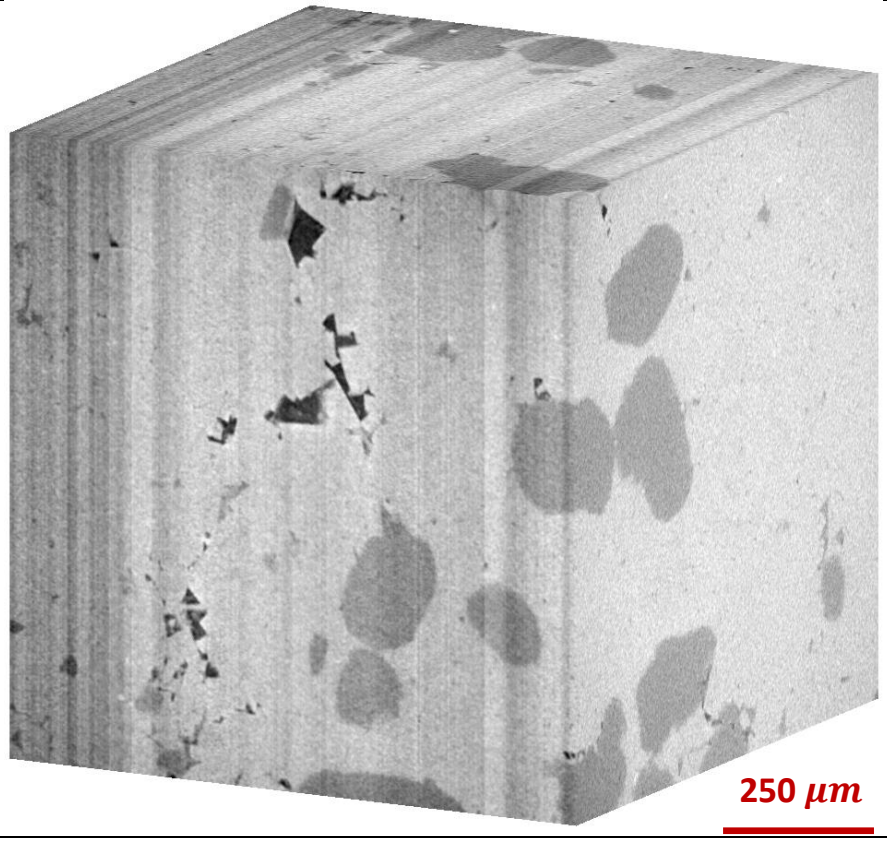
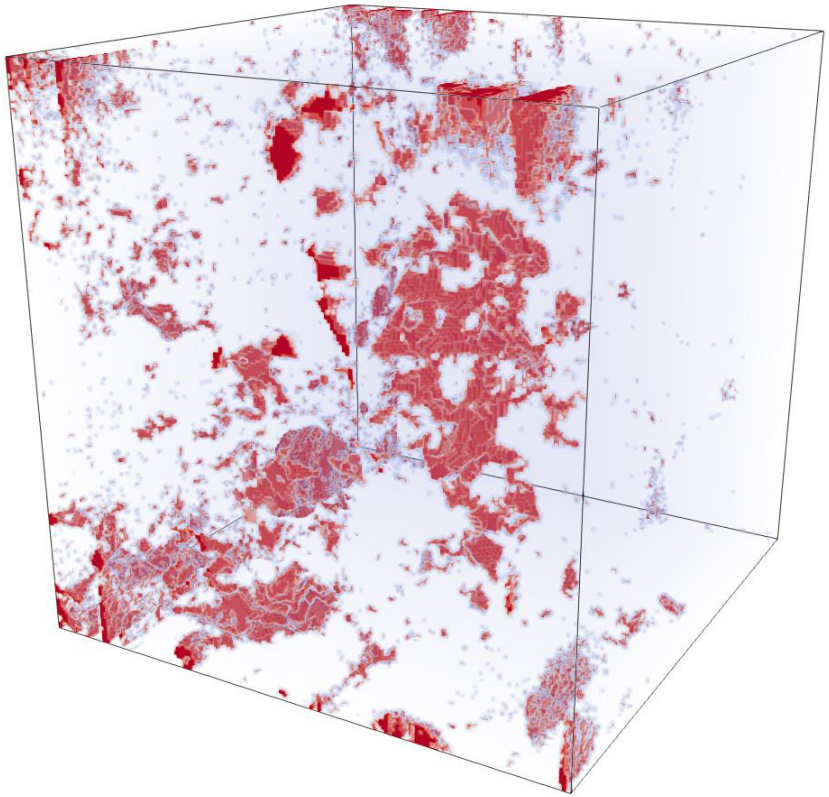
This table here includes all petrophysical properties that measured or calculated in this work. For convenience of comparison, we repeat the NMR porosity and T_{2ML} here. The surface to volume ratio S_{por} derived from BET test, and transverse surface relaxivity ρ_2 are calculated from S_{por}. Since only two of granstones, one of both packstones and mudstones have BET S_{por} available, we used average S_{por} to calculated ρ_2. NMR S_{por} therefore can be calculated from ρ_2 and T_{2ML} data. All S_{por} analyzed in this work are NMR S_{por}. Characteristic pore size Λ is calculated from T_{2ML} using Equation 3. Other petrophysical properties that discussed and listed here are tortuosity τ, formation factor F and cementation factor m which are all unitless parameters. Permeabilities calculated from K-C eqation are also listed here to compare with air-permeability.

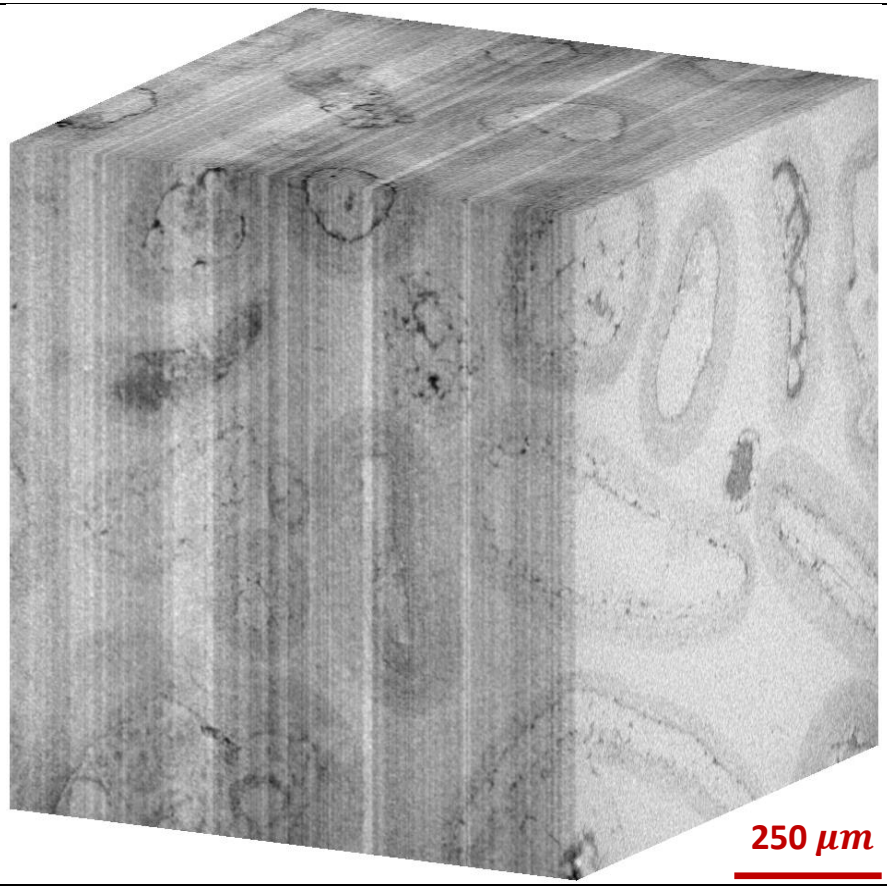
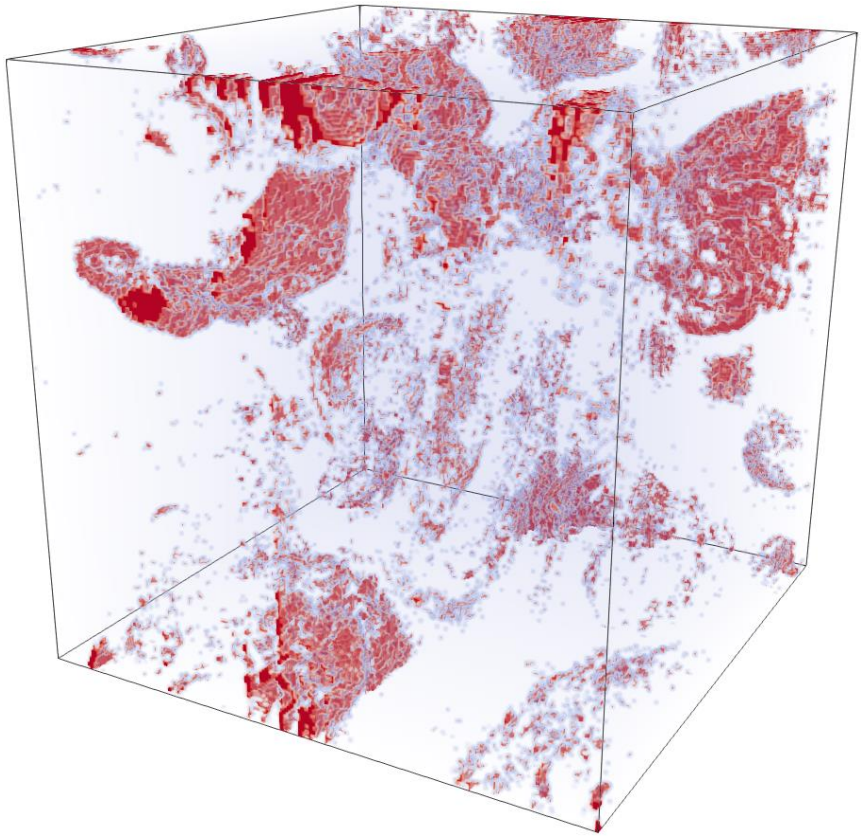
Lithofacies	**Sample I.D.**	**NMR Porosity**	T_{2ml}	S_{por} **(BET)**	ρ_2 **(BET)**	S_{por} **(NMR)**	Λ	τ	F	m	**Permeiability K-C equation**
		(%)	(ms)	(/μm)	(μm/s)	(/μm)	(nm)				(mD)
Grainstones	3	21.03	253.46		2.57	1.32	1267.28	3.29	53.44	2.49	5.59
	9	17.22	173.01		2.24	2.89	865.04	2.81	51.81	2.10	1.37
	11	19.42	263.16		2.95	1.27	1315.82	2.61	35.34	2.17	8.21
	16	23.88	491.83	35.05	4.21	0.12	2459.13	3.71	54.35	2.91	51.43
	27	9.93	225.5		5.49	1.48	1127.51	2.18	68.92	1.58	0.27
	37	20.17	457.6		4.88	0.44	2288.00	3.97	75.19	2.77	15.84
	38	18.27	274.73		3.32	1.21	1373.67	2.49	37.47	2.02	2.84
	39	22.47	229.71		2.12	2.18	1148.56	1.73	11.80	1.80	33.62
	44	18.8	360.63		4.18	0.69	1803.14	2.28	25.89	2.02	32.67
	51	8.54	72.852		2.11	13.73	364.26	3.32	119.65	2.01	0.08
	51b	5.92	37.397		2.36	53.48	186.99	4.79	358.51	2.14	0.01
	53	15.43	159.77		2.76	6.26	798.87	2.81	48.32	2.14	2.65
	57	19.87	254.9		2.12	1.96	1274.51	2.59	31.54	2.23	11.25
	59	10.21	89.013		3.37	11.23	445.07	2.47	95.49	1.66	0.00
	64	25.89	438.26		2.45	0.46	2191.28	2.80	29.10	2.57	54.59
	65	19.51	221.28		2.78	2.26	1106.42	2.64	35.21	2.20	6.70
	66	19.81	255.06		2.33	1.96	1275.28	2.68	36.04	2.22	6.14
	67	19	203.09		0.31	2.46	1015.44	1.55	13.54	1.51	9.54

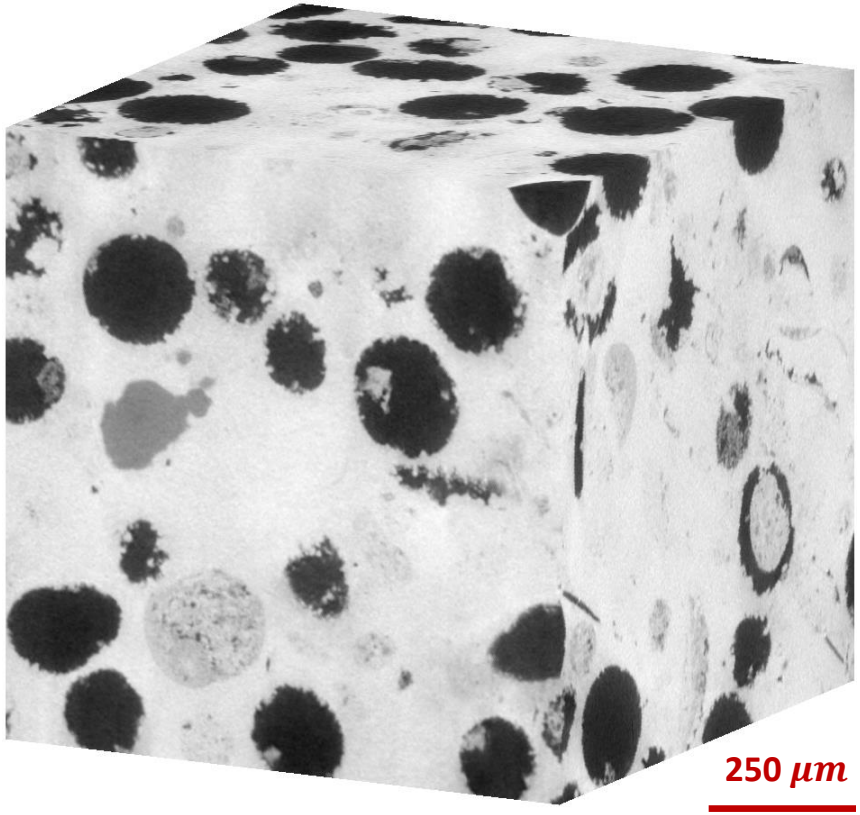
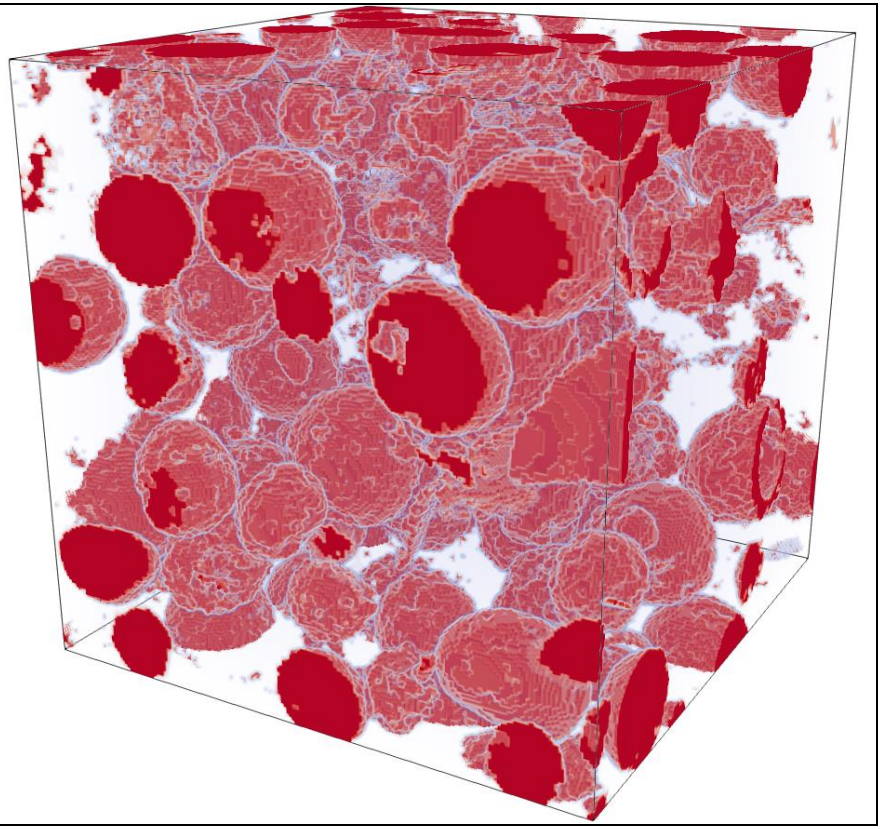
	69	4.18	5.066		5.41	197.39	25.33	2.26	181.15	1.46	0.00
	70	7.75	168.11		1.67	2.97	840.54	3.11	105.32	1.95	0.91
	71	6.04	40.603		0.59	49.26	203.02	2.30	126.61	1.52	0.00
	72	5.32	12.23		2.33	163.53	61.15	2.89	112.59	1.86	0.01
	81	25.37	292.83		3.09	1.14	1464.17	2.35	21.08	2.28	35.92
	82	17.85	247.84		2.67	2.02	1239.21	2.44	35.16	2.00	3.85
	85	24.19	315.14		3.01	1.06	1575.72	2.50	25.66	2.30	28.40
	86	21.56	306.14		0.40	1.09	1530.72	2.52	27.09	2.27	23.40
	89	6.88	11.047	48.22	3.72	169.92	55.24	3.22	155.04	1.87	0.00
	92	30.7	612.64		2.34	0.27	3063.22	3.46	35.79	3.27	141.82
	101	23.17	261.94		4.94	1.27	1309.71	3.34	44.98	2.73	11.26
	109b	21.76	398.22		3.74	0.63	1991.11	2.71	32.64	2.34	30.06
	115	30.94	816.07		3.17	0.20	4080.36	4.46	62.08	3.63	118.31
	123	18.74	317.32		1.50	1.05	1586.58	3.01	43.53	2.40	8.25
	130	17.6	250		0.83	2.00	1249.99	2.39	45.88	1.83	1.20
	135	8.4	50.583		3.26	39.54	252.92	2.92	122.81	1.81	0.01
	145	5.79	18.229		5.07	109.72	91.14	2.50	106.87	1.64	0.00
	151	20.3	305.55		3.34	1.09	1527.76	2.96	41.65	2.39	11.26
	154	14.9	330.24		1.00	1.01	1651.20	4.17	116.60	2.50	2.52
	156	4.4	56.34		3.20	35.50	281.70	2.80	195.82	1.64	0.00
	157	7.38	30.134		3.56	66.37	150.67	2.36	79.37	1.65	0.02
	164	23.78	367.58		3.86	0.91	1837.91	3.42	45.07	2.82	26.12
	165	20.7	343.25		1.61	0.73	1716.26	3.06	42.48	2.48	16.03
Packstones	13-18	7.4	109.05	0.36	0.69	13.20	54.52	3.59	174.34	1.98	37.35
	13-46	5.304	171.96		0.31	8.37	85.98	6.14	710.44	2.24	11.71
	14-9	8.75	124.94		0.79	11.52	62.47	3.08	108.40	1.92	110.25
	18-30	4.75	99.231		0.61	14.51	49.62	2.81	166.71	1.68	13.33
	18-33	5.81	101.81		0.58	14.14	50.91	2.71	126.87	1.70	27.58
Mudstones	16-15	6.63	34.866	2.87	0.23	123.50	17.43	2.52	95.99	1.68	6.52
	20-37	2.25	25.569		0.08	168.41	12.78	1.44	91.71	1.19	0.71
	30-4	3.76	23.177		0.16	185.79	11.59	3.10	255.08	1.69	0.49
	30-24	3.03	19.03		0.16	226.27	9.52	2.96	288.54	1.62	0.19
	33-25	12.94	253.69		0.08	16.97	126.84	1.96	29.63	1.66	2885.85

Appendix C – CT images and digital rocks

Sample I.D.	CT image (black is pore)	Digital rock (red is pore)
W33-25		

W13-18		

LKC89		

LKC92		

LKC135	